

Bayes rule ~~various~~ variants

Note: in the discrete case,
density function
 $f_Y(y) = P(Y=y)$

① θ discrete, X discrete

$$f_{\theta|X}(\theta|x) = P(\theta=\theta|x) = \frac{P(\theta=\theta) \cdot P(X=x|\theta)}{\sum_{\theta'} P(\theta=\theta') \cdot P(X=x|\theta')}$$

examples: 6.8-1; problem 2 from practice midterm

② θ discrete, X continuous

$$f_{\theta|X}(\theta|x) = P(\theta=\theta|x) = \frac{P(\theta=\theta) \cdot f_{X|\theta}(x|\theta)}{\sum_{\theta'} P(\theta=\theta') \cdot f_{X|\theta}(x|\theta')}$$

examples: 6.8-1 - change Poisson to Exponential distribution

③ θ continuous, X discrete

$$f_{\theta|X}(\theta|x) = \frac{f_{\theta}(\theta) P(X=x|\theta)}{\int P(X=x|\theta') \cdot f_{\theta}(\theta') d\theta'}$$

examples: $\theta \sim \text{Unif}[0,1]$, $X = \begin{cases} 4 & \text{with probability } \theta \\ 3 & \text{with probability } 1-\theta \end{cases}$

④ θ continuous, X continuous

$$f_{\theta|X}(\theta|x) = \frac{f_{\theta}(\theta) \cdot f_{X|\theta}(x|\theta)}{\int f_{\theta}(\theta') f_{X|\theta}(x|\theta') d\theta'}$$

examples: gaussian example from class (see attachment to Jan 28 lecture), Problem 1 hw4.